EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 04

Vectors

Components, Dot Product and Practice

By - Excellence Programme

3 October 2026

Topics to be covered

01 Adding by components

02 Change in velocity and rotation

03 Dot product

04 Unit vector practice

05 Angle and direction from components

06 Fresh JEE-style practice

07 Cross product (added)

What we are skipping today

Already done in Vector 01 and Vector 02 before slide 91:

· scalar multiply, compass language, reverse/scale angles

· rectangular components and magnitude

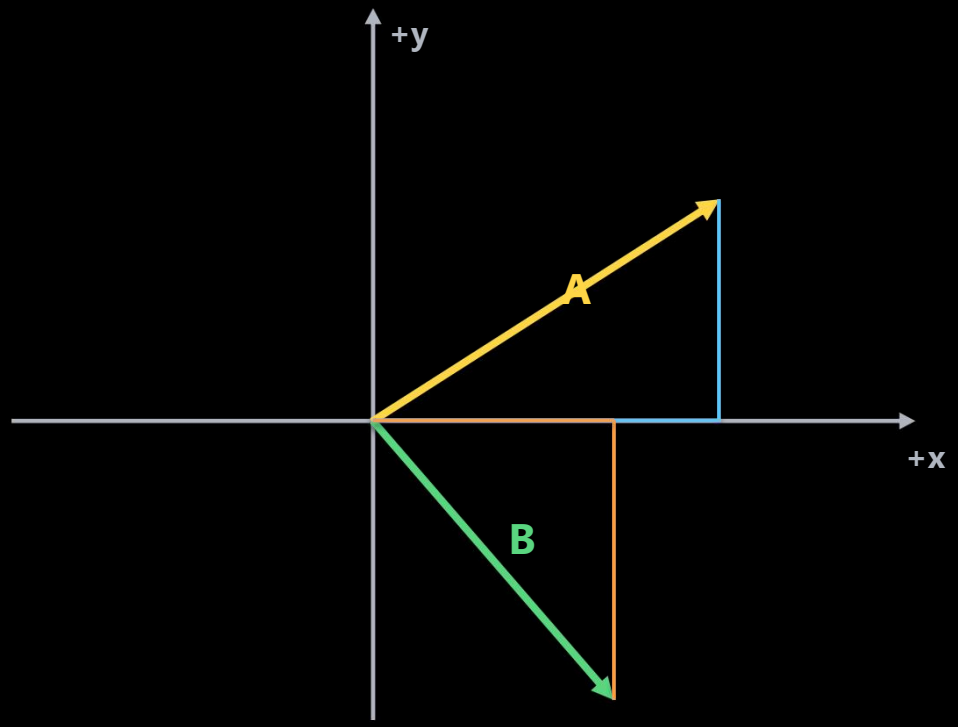
· triangle law, resultant formula, range, equal-magnitude tricks

· subtraction idea, and the repeated resultant drills

Today starts at adding by components, then moves into dot product and new practice. Cross product is added at the end.

Adding by components is adding columns

The x-parts combine with x-parts. The y-parts combine with y-parts. This is the cleanest method for many vectors.



$$\mathbf{A + B = (A_x + B_x)\,\hat{i} + (A_y + B_y)\,\hat{j}}$$

Column idea

x-column decides left-right. y-column decides up-down.

3D same rule

$A + B = (A_x+B_x)\hat{i} + (A_y+B_y)\hat{j} + (A_z+B_z)\hat{k}$

Add by components

$A = 4\,\hat{i} + 3\,\hat{j}$ and $B = 3\,\hat{i} - 4\,\hat{j}$. Find $A + B$, $A - B$, and $|A| + |B|$.

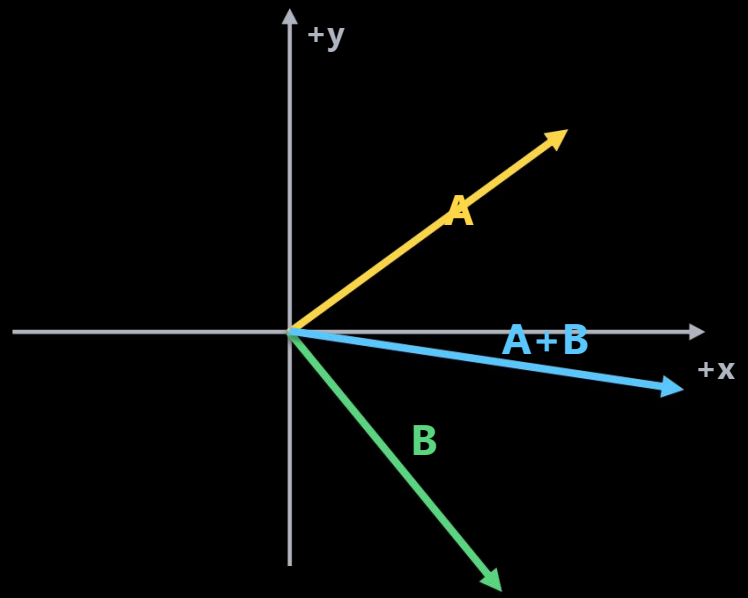
Solution: Add by components

Step 1: $\vec{A} + \vec{B} = 7\,\hat{i} - 1\,\hat{j}$

Step 2: $\vec{A} - \vec{B} = 1\,\hat{i} + 7\,\hat{j}$

Step 3: $|\vec{A}| = 5$ and $|\vec{B}| = 5$, so $|\vec{A}| + |\vec{B}| = 10$

Sum of magnitudes is 10. Magnitude of the sum is √50.



Rough work / extra work

Change in velocity

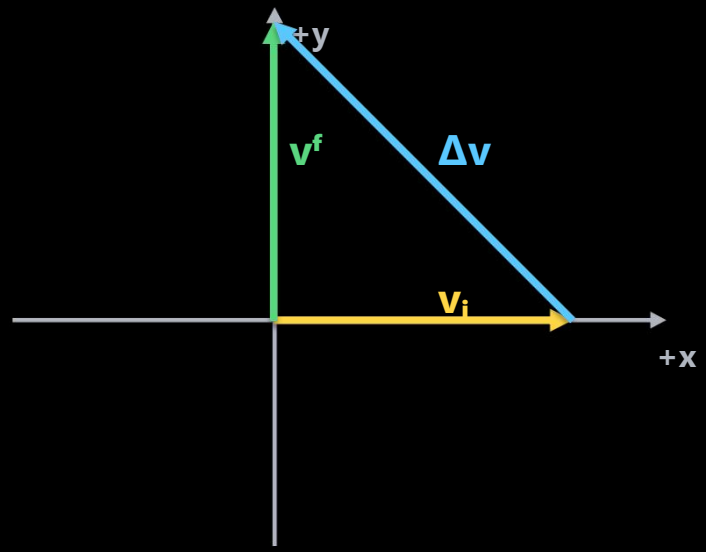
Velocity changes from 10 m/s east to 10 m/s north. Find the change in velocity.

Solution: Change in velocity

Step 1: Change = final - initial = 10 ĵ - 10 î

Step 2: Magnitude = √(100 + 100) = 10 √2

10 √2 m/s, pointing north of west.



Rotate 2 î

A = 2 î is rotated by 90°, 180°, 270° and 360° anticlockwise. Write each new vector.

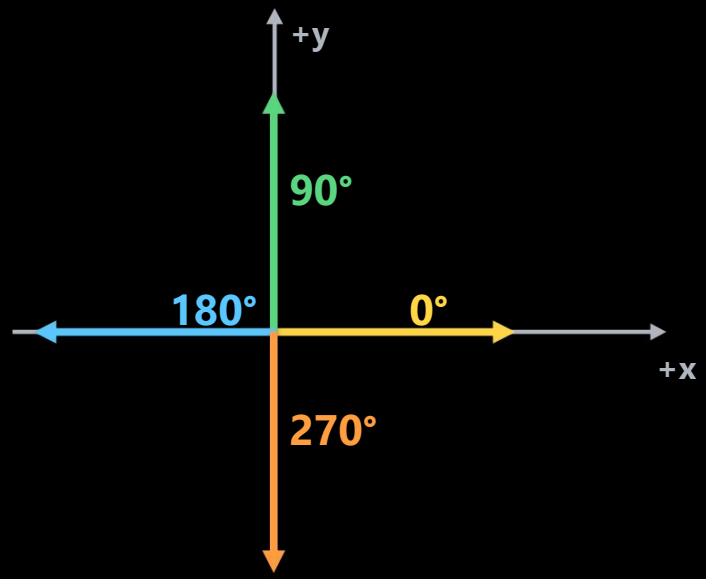
Solution: Rotate 2 î

Step 1: 90°: 2 ĵ

Step 2: 180°: -2 î

Step 3: 270°: -2 ĵ. 360°: 2 î again

Rotation keeps the magnitude and changes only the direction.



Rough work / extra work

Dot product

$$\mathbf{A} \cdot \mathbf{B} = A\,B\cos\theta = A_x B_x + A_y B_y + A_z\,B_z$$

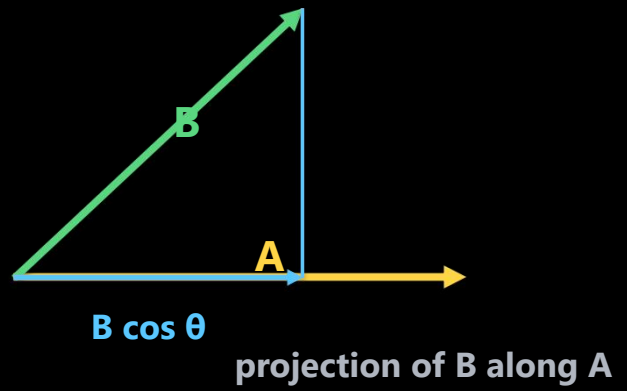
If the dot product is 0, the vectors are perpendicular.

cos θ = (A · B) / (A B). That is how the angle is found from components.

Component of B along A = (A · B) / A.

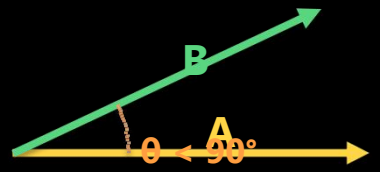
$\hat{i} \cdot \hat{i} = 1$. $\hat{i} \cdot \hat{j} = 0$. The same rule holds for $\hat{j}$ and $\hat{k}$.

A unit vector dotted with itself is 1.

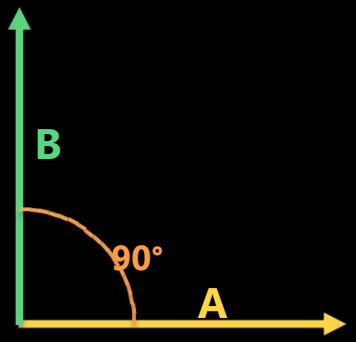


Dot product tells alignment

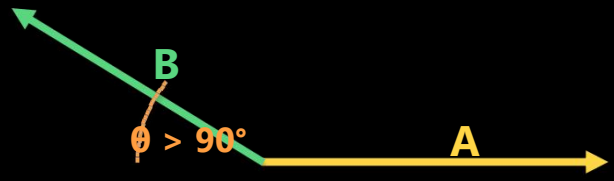
Dot product is positive when vectors face mostly the same way, zero at 90°, and negative when they face opposite way.



$A \cdot B > 0$



$A \cdot B = 0$



$A \cdot B < 0$

Why this matters

It quickly tests perpendicular vectors and helps find the angle between two vectors.

Are these two vectors perpendicular?

A = 3 î + 4 ĵ + 5 k̂ and B = 3 î + 4 ĵ - 5 k̂. What is the angle between them?

Solution: Are these two vectors perpendicular?

Step 1: $A \cdot B = 9 + 16 - 25 = 0$

The angle is 90°.

Angle with 4 $\hat{i}$

$A = 2\hat{i} + 6\hat{j} + 3\hat{k}$ and $B = 4\hat{i}$. Find $\cos\theta$.

Solution: Angle with 4 î

Step 1: $A \cdot B = 8$

Step 2: $|A| = \sqrt{(4 + 36 + 9)} = 7$ and $|B| = 4$

Step 3: $\cos \theta = 8 / 28 = 2/7$

$\theta = \cos^{-1}$ of 2/7.

Rough work / extra work

Unit vectors from components

$\hat{i}$ itself is already a unit vector.

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$.

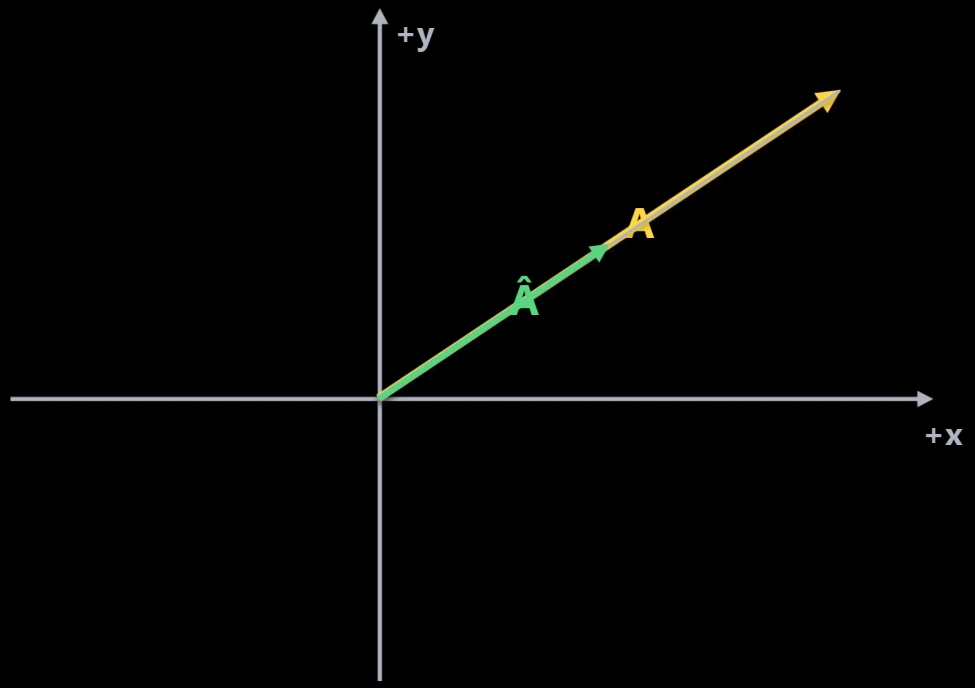
Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$.

$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1, because $\sin^2 + \cos^2 = 1$.

Unit vector of $3\,\hat{i} + 4\,\hat{j}$ is $(3\,\hat{i} + 4\,\hat{j}) / 5$.

Unit vector keeps direction and removes size

A unit vector is only the direction of a vector. It has length 1 and no physical unit.



$$\hat{A} = A / |A|$$

Example

For $A = 3\hat{i} + 4\hat{j}$, $|A| = 5$, so $\hat{A} = (3\hat{i} + 4\hat{j})/5$.

Do not attach N, m/s or kg to a unit vector.

Perpendicular forces A and A/2

Two forces of magnitudes A and A/2 are perpendicular. Find the magnitude of their resultant.

Solution: Perpendicular forces A and A/2

Step 1: cos 90° = 0, so $R = \sqrt{(A^2 + A^2/4)}$

Step 2: $R = \sqrt{(5\,A^2 / 4)} = (A/2)\,\sqrt{5}$

$(\sqrt{5} / 2) \times A$.

Rough work / extra work

Resultant is half the larger vector

The resultant of two vectors is half the larger one, and it is perpendicular to the smaller one. Find the angle between the vectors.

Solution: Resultant is half the larger vector

Step 1: R perpendicular to the smaller vector means $(A + B) \cdot B = 0$

Step 2: $\cos\theta = -B/A$

Step 3: $R = A/2$ then gives $B = (\sqrt{3}/2)\,A$

$\cos\theta = -\sqrt{3}/2$, so $\theta = 150°$.

Acceleration of a 4 kg body

$F_1 = 5\,\hat{i} + 8\,\hat{j} + 7\,\hat{k}$ and $F_2 = 3\,\hat{i} - 4\,\hat{j} - 3\,\hat{k}$ act on 4 kg. Find the acceleration.

Solution: Acceleration of a 4 kg body

Step 1: Net force = $8\,\hat{i} + 4\,\hat{j} + 4\,\hat{k}$

Step 2: a = net force / 4

$a = 2\,\hat{i} + 1\,\hat{j} + 1\,\hat{k}$.

Angle conditions to lock

Condition	Angle
$\|A\| = \|B\| = \|A + B\|$	120°
$\|A\| = \|B\| = \|A - B\|$	60°
$\|A + B\| = \|A - B\|$	90°
$\|A + B\| = \|A\| + \|B\|$	0°
$\|A - B\| = \|A\| + \|B\|$	180°
$\|A + B\| = \|A\| - \|B\|$	180°
$\|A - B\| = \|A\| - \|B\|$	0°

Rough work / extra work

$A = 6\,\hat{i}$ and $B = 8\,\hat{j}$

Find A + B, A - B, |A| + |B|, and |A + B|.

Solution: A = 6 î and B = 8 ĵ

Step 1: A + B = 6 î + 8 ĵ

Step 2: A - B = 6 î - 8 ĵ

Step 3: |A| + |B| = 6 + 8 = 14, and |A + B| = √(36 + 64) = 10

Sum of lengths is 14. Length of the sum is 10.

Direction of A + B from A

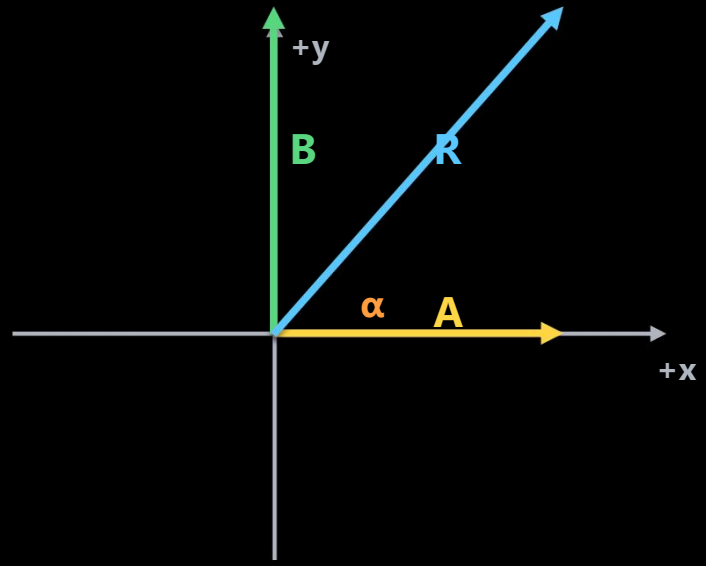
A = 6 î and B = 8 ĵ. Find the direction of A + B from A.

Solution: Direction of A + B from A

Step 1: $\vec{A} + \vec{B} = 6\,\hat{i} + 8\,\hat{j}$

Step 2: $\tan\alpha = 8 / 6 = 4 / 3$

$\alpha = \tan^{-1}$ of 4/3 from A.



Unit vector list

$\hat{i}$ itself is already a unit vector.

Unit vector of $\hat{i} + \hat{j}$ is $(\hat{i} + \hat{j}) / \sqrt{2}$.

Unit vector of $\hat{i} + \hat{j} - \hat{k}$ is $(\hat{i} + \hat{j} - \hat{k}) / \sqrt{3}$.

$\sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ already has magnitude 1.

Unit vector of $3\hat{i} + 4\hat{j}$ is $(3\hat{i} + 4\hat{j}) / 5$.

$$\hat{A} = A / |A|$$

Rough work / extra work

Find β for a unit vector

A = 0.4 î + 0.3 ĵ + β k̂ is a unit vector. Find β.

Solution: Find β for a unit vector

Step 1: For a unit vector, $|A| = 1$

Step 2: $\sqrt{(0.16 + 0.09 + \beta^2)} = 1$

Step 3: $0.25 + \beta^2 = 1$, so $\beta^2 = 0.75$, $\beta = \pm\sqrt{3} / 2$

$\beta = \pm (\sqrt{3})/2$.

Find α if A is perpendicular to B

A = 2 î + 3 ĵ + α k̂ and B = î - 2 ĵ + 4 k̂. If A is perpendicular to B, find α.

Solution: Find α if A is perpendicular to B

Step 1: $A \cdot B = 0$

Step 2: $2 - 6 + 4\alpha = 0$

Step 3: $4\alpha = 4$, so $\alpha = 1$

$\alpha = 1$.

Angle of a vector from the x-axis

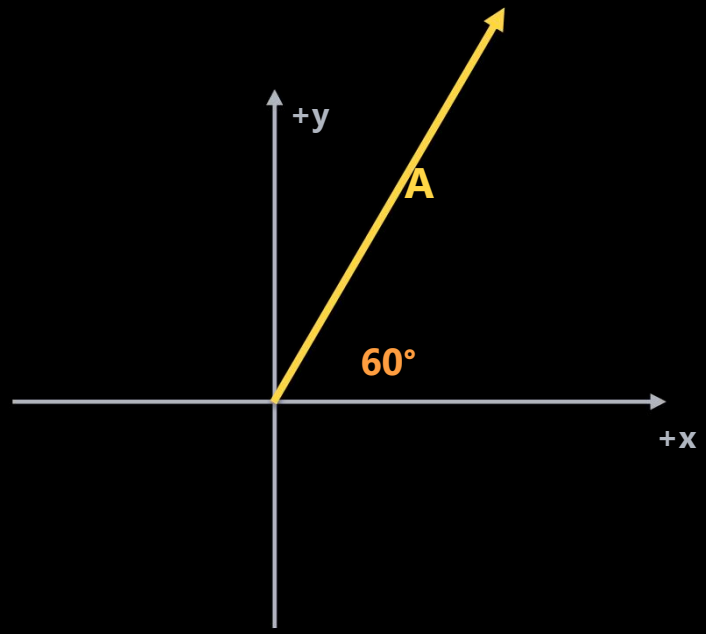
A = 5 î + 5 √3 ĵ. Find the angle of A from the x-axis.

Solution: Angle of a vector from the x-axis

Step 1: $\tan \alpha = A_y / A_x = 5\sqrt{3} / 5 = \sqrt{3}$

Step 2: $\alpha = 60°$

60° from the +x axis.



Rough work / extra work

Find the smaller vector

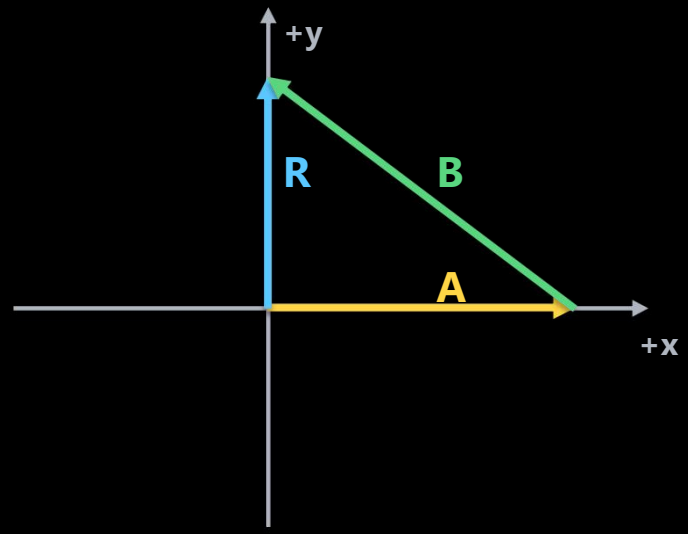
Resultant is perpendicular to the smaller vector and half the larger vector. Find the smaller magnitude in terms of the larger B.

Solution: Find the smaller vector

Step 1: Right triangle: $A^2 + (B/2)^2 = B^2$

Step 2: $A^2 = B^2 - B^2/4 = 3B^2/4$

$A = (\sqrt{3} / 2)\ B.$



Three equal vectors OP, OQ, OR

Three vectors each of magnitude A act as OP along +x, OQ along +y, and OR 45° below +x. The resultant is A √(x). Find x.

Solution: Three equal vectors OP, OQ, OR

Step 1: OP + OQ = $A\sqrt{2}$ at 45° above +x

Step 2: OR is 45° below +x, so that intermediate resultant and OR are at 90°

Step 3: $R = \sqrt{((A\sqrt{2})^2 + A^2)} = A\sqrt{3}$

x = 3.

Rough work / extra work

Vector at 30° from the y-axis

A vector in the x-y plane makes 30° with the y-axis. Its y-component is 2 √3. Find the x-component.

Solution: Vector at 30° from the y-axis

Step 1: $A_y = A \cos 30° = 2\sqrt{3}$

Step 2: $A(\sqrt{3}/2) = 2\sqrt{3}$, so $A = 4$

Step 3: $A_x = A \sin 30° = 4 \times 1/2 = 2$

x-component = 2.

Cross product — the easy picture

Dot product gave a number. Cross product gives a new vector.

A × B is perpendicular to both A and B

Think: A and B lie on a table. A × B stands straight up from the table.

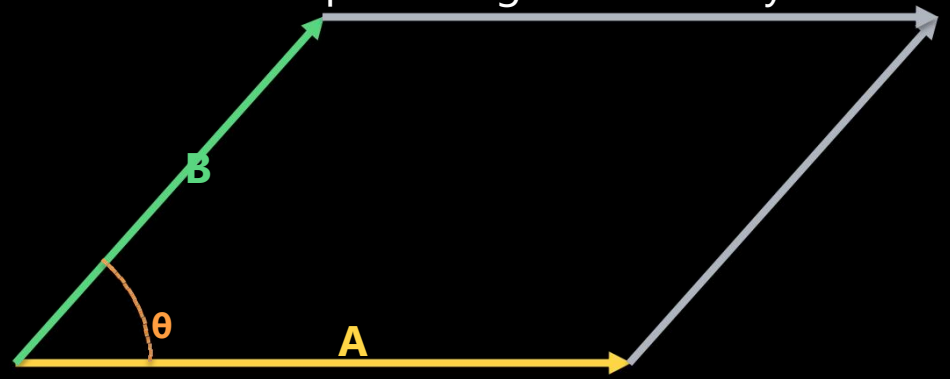
If A and B are parallel, A × B = 0. There is no unique upright direction.

Magnitude first: A B sin θ

Only the turning part of B counts. That part is B sin θ.

$$|A \times B| = A\,B \sin\theta$$

This is also the area of the parallelogram made by A and B.



Memory

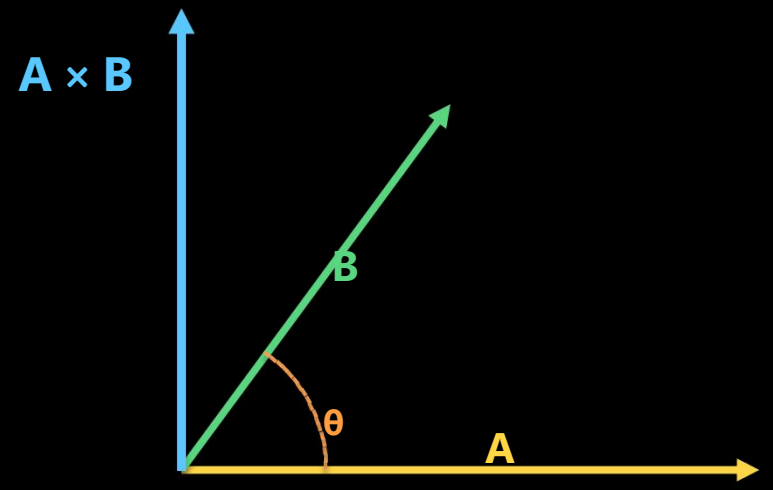
sin for cross, cos for dot.

Max / zero

Max at 90°. Zero at 0° or 180°.

Direction: right-hand rule

Point fingers along A. Curl them toward B through the smaller angle. Thumb shows A × B.



$$A \times B = -(B \times A)$$

Order matters

Swap the two arrows and the result flips.

Curl through the smaller angle between A and B.

Rough work / extra work

Dot vs cross — one glance

	Dot product	Cross product
Result	a number (scalar)	a vector
Formula size	$A\,B\cos\theta$	$A\,B\sin\theta$
Zero when	perpendicular (90°)	parallel (0° or 180°)
Order	$A \cdot B = B \cdot A$	$A \times B = -B \times A$

Same pair of arrows. Cos asks 'how parallel?'. Sin asks 'how perpendicular?'.

Unit-vector cycle (easy)

$$\hat{\imath} \times \hat{\jmath} = \hat{k},\ \hat{\jmath} \times \hat{k} = \hat{\imath},\ \hat{k} \times \hat{\imath} = \hat{\jmath}$$

Go forward in the cycle i → j → k → i: answer is positive.

Go backward: answer gets a minus sign.

$$\hat{\imath} \times \hat{\imath} = 0,\ \hat{\jmath} \times \hat{\jmath} = 0,\ \hat{k} \times \hat{k} = 0$$

Same letter crossed with itself is always zero.

Component formula without fear

Write the 3 × 3 pattern once. Read the three rows for $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$.

$$\mathbf{A \times B = (A_y B_z - A_z B_y)\,\hat{\imath} - (A_x B_z - A_z B_x)\,\hat{\jmath} + (A_x B_y - A_y B_x)\,\hat{k}}$$

Shortcut for xy-plane: if $A = A_x\,\hat{\imath} + A_y\,\hat{\jmath}$ and $B = B_x\,\hat{\imath} + B_y\,\hat{\jmath}$,

$$\mathbf{A \times B = (A_x B_y - A_y B_x)\,\hat{k}}$$

In the plane, the cross product points along $\pm\hat{k}$ only.

Find |A × B|

A = 3 and B = 4. Angle between them is 30°. Find |A × B|.

Solution: Find |A × B|

Step 1: $|A \times B| = A\,B \sin\theta = 3 \times 4 \times \sin 30°$

Step 2: $\sin 30° = 1/2$, so $|A \times B| = 6$

$|A \times B| = 6$.

Rough work / extra work

$\hat{\imath} \times \hat{\jmath}$ and $\hat{\jmath} \times \hat{\imath}$

Find $\hat{\imath} \times \hat{\jmath}$ and $\hat{\jmath} \times \hat{\imath}$.

Solution: $\hat{i} \times \hat{j}$ and $\hat{j} \times \hat{i}$

Step 1: $\hat{i} \times \hat{j} = \hat{k}$

Step 2: $\hat{j} \times \hat{i} = -(\hat{i} \times \hat{j}) = -\hat{k}$

$\hat{k}$ and $-\hat{k}$. Order flips the sign.

Cross product in the xy-plane

$\mathbf{A} = 2\,\hat{\imath} + 3\,\hat{\jmath}$ and $\mathbf{B} = 4\,\hat{\imath} - \hat{\jmath}$. Find $\mathbf{A} \times \mathbf{B}$.

Solution: Cross product in the xy-plane

Step 1: $\vec{A} \times \vec{B} = (A_x B_y - A_y B_x)\,\hat{k}$

Step 2: $= (2)(-1) - (3)(4) = -2 - 12 = -14$

$\vec{A} \times \vec{B} = -14\,\hat{k}$.

When is A × B zero?

A = 2 î + 4 ĵ and B = î + 2 ĵ. Find A × B. What does the answer mean?

Solution: When is A × B zero?

Step 1: $\mathbf{A} \times \mathbf{B} = (2 \times 2 - 4 \times 1)\,\hat{k} = (4 - 4)\,\hat{k} = 0$

Step 2: $\mathbf{B} = (1/2)\,\mathbf{A}$, so the arrows are parallel

$\mathbf{A} \times \mathbf{B} = 0$. Parallel vectors give zero cross product.

Rough work / extra work

Unit vector along A × B

A = î + ĵ and B = î − ĵ. Find a unit vector along A × B.

Solution: Unit vector along A × B

Step 1: $\mathbf{A} \times \mathbf{B} = (1\times(-1) - 1\times1)\,\hat{k} = -2\,\hat{k}$

Step 2: $|\mathbf{A} \times \mathbf{B}| = 2$, so unit vector $= (-2\,\hat{k})/2 = -\,\hat{k}$

Unit vector $= -\,\hat{k}$.

Vector 04 lock lines

Add by columns: (Ax + Bx) î + (Ay + By) ĵ

Delta v = v_f - v_i. Same speed can still give large Delta v.

A · B = A B cos θ. Unit vector = vector / magnitude.

|A × B| = A B sin θ. Right-hand rule. A × B = − B × A.

In the xy-plane: A × B = (Ax By − Ay Bx) k̂.

Practice

1. Redo $A = 4\,\hat{i} + 3\,\hat{j}$ and $B = 3\,\hat{i} - 4\,\hat{j}$ for sum and difference.

2. East→north at 10 m/s: write Δv and $|\Delta v|$.

3. Find β if $0.4\hat{i} + 0.3\hat{j} + \beta\,\hat{k}$ is a unit vector.

4. Redo the OP, OQ, OR resultant question.

5. Find $A \times B$ for $A = 3\,\hat{i} + \hat{j}$ and $B = 2\,\hat{i} + 4\,\hat{j}$.

6. Write $\hat{i} \times \hat{j}$, $\hat{j} \times \hat{k}$, and $\hat{j} \times \hat{i}$.

Dot uses cos. Cross uses sin. Do not mix them.

Rough work / extra work